

1. PROJECT & TEAM INFORMATION

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models
- **Team ID:** SWTID-2026-7649
- **Team Leader:** N.Sopya
- **Team Members:** S S Sathyabala, Mohamed Ajmal, Mohamed Irfan P, Mohamed jubair.P

2. PERSONA OVERVIEW

- **User Persona:** Creative enthusiasts, amateur writers, and students who lack illustration skills but possess strong imaginative storytelling capabilities.
- **Core Goal:** To instantly create a visually appealing comic book from a basic text-based story idea, completely bypassing the need for manual drawing or graphic design software.

3. CUSTOMER JOURNEY STAGES

Phase 1: Discovery & Entry

- **User Action:** The user navigates to the ComicCraft web application URL.
- **Touchpoint:** Web Browser, Homepage (index.html).
- **User Emotion:** Curious and slightly overwhelmed by the blank canvas of a new tool.
- **Pain Point:** Unsure of how detailed the story prompt needs to be for the AI to understand.
- **System Opportunity:** Provide clear placeholder text and simple dropdown menus for Setting, Tone, and Art Style to guide the user's input effortlessly.

Phase 2: Input & Interaction

- **User Action:** The user types their story prompt, enters the main character's name, and selects preferences (e.g., "Forest", "Funny", "Realistic"). Clicks the "Generate Comic" button.

- **Touchpoint:** HTML Form, FastAPI POST Route (/generate).
- **User Emotion:** Excited, anticipatory, and hopeful.
- **Pain Point:** The waiting time for complex AI image generation can feel extensive if no feedback is given.
- **System Opportunity:** Implement a loading state or spinner on the frontend to manage wait-time expectations while the backend processes the request.

Phase 3: Generation & Processing (Behind the Scenes)

- **System Action:** FastAPI routes the prompt to Gemini Flash for the panel outline, Gemini Pro for the narrative story, and Hugging Face Stable Diffusion for generating the illustrations.
- **Touchpoint:** Backend AI APIs and internal Python routing.
- **System Goal:** Ensure seamless fallback mechanisms (e.g., dual-engine API switching) if an external API times out or rate-limits the IP address.

Phase 4: Review & Engagement

- **User Action:** The user is seamlessly redirected to the preview page to read the generated comic panel by panel.
- **Touchpoint:** Comic Preview Page (comic_preview.html).
- **User Emotion:** Delighted and amazed at the accurate visual representation of their textual input.
- **System Opportunity:** Display panels clearly with prominent titles, high-quality images, and segregated narration text for a comfortable reading experience.

Phase 5: Export & Completion

- **User Action:** The user clicks the "Download Your Comic as PDF" button.
- **Touchpoint:** Export Route, Success Page (export_success.html).
- **User Emotion:** Highly satisfied and accomplished.

- **System Opportunity:** Provide a beautifully formatted PDF utilizing the FPDF library and include a "Go Create Another Comic" button to encourage platform retention.